# COLO829: somatic mutation and RNA evidence

Whole source-call audit | matched-normal DNA baseline | original tumor RNA | pinned reference ORFs. Counts below describe the specified public call sets and RNA run, not every biological mutation or a validated full-length isoform.

## At a glance

| Evidence layer | Count | Interpretation |
|---|---|---|
| Ingested small variants | 46,064 | 44,005 SNVs; 2,059 indels |
| Ingested SV source records | 188 | DNA breakpoint records/events, counted in their source sets |
| Ensembl 87 coding effect | 373 | Any annotated protein-coding transcript, reference model |
| orfdata ncORF coding-block overlap | 5 | Candidate reference ORF overlap; distinct from mutant translation |
| Original RNA ALT-observed calls | 1,933 | 35,949 ALT reads; 35,617 ALT fragments |
| Calls with local mutant protein contexts | 124 | 12,218 ranked contexts; full-length isoforms unresolved |

## Coding annotation and its denominator

Of 46,064 DNA small-variant records, 373 change the coding sequence of at least one Ensembl 87 transcript. The pinned orfdata 0.34.1 GRCh38 catalog (7,264 reference ncORF models; release 2026-10-11.1) has exact coding-block overlaps at 5 calls. The groups overlap at 0 calls, so the union is 378. These are annotation opportunities. The catalog does not assess a cell-line-specific mutant ORF, RNA expression, or peptide evidence.

| Reference annotation | All DNA calls | With original RNA ALT |
|---|---|---|
| Ensembl coding effect | 373 | 180 |
| orfdata coding-block overlap | 5 | 2 |
| Union of either category | 378 | see per-call table |

SNVs query their changed reference base. Deletions query removed reference bases; insertions use the VCF anchor as a candidate overlap. The per-call table preserves that rule. A coding-block hit does not prove the inserted sequence is in frame.

## Original RNA and local protein evidence

RNA assay: SRR8615617, Illumina paired RNA. Each nominated small variant has a status from this run. An ALT read is a sequence observation; it is not a full-length mutant isoform.

| RNA allele status | Small-variant records |
|---|---|
| alternate observed | 1,933 |
| reference only | 1,637 |
| reference and other only | 36 |
| other allele only | 27 |
| alignment without allele call | 166 |
| no alignment overlap | 42,265 |

The 1,933 ALT-observed calls contribute 35,949 allele-read and 35,617 fragment observations. Isovar resolves at least one read-compatible **local** mutant protein context for 124 calls (12,218 ranked rows). The remaining 1,809 ALT-observed calls have no local translation under the stated method. No-overlap and uncallable statuses cannot establish absence of RNA expression.

## High-depth indel and polished sequence comparison

At CDKN2A chr9:21971154:CCG>C, all 6,984 ALT allele reads were evaluated. The top local frameshift context is **MMMGSARVAELLLLHGGAQLRRPR** with 5,338 supporting reads. Its 1,416 ranked local contexts reflect read-spanning alternatives; none is asserted to be a full-length isoform.

TAGET polished representatives are an independent culture and assay: 132 SMaHT SNVs have at least one tumor-positive representative carrying ALT. The 0/1 representative-sequence observations and full-length cluster counts are separate from original allele-spanning read support. The polished count table has 51,387 tumor-positive transcript IDs, but those counts do not measure mutant allele support.

## Structural-variant RNA screen

| Breakend screen outcome | Source records |
|---|---|
| No breakend-window alignment | 78 |
| One end covered; no split candidate | 36 |
| Both ends covered; no split candidate | 57 |
| Cross-end mate context; path unresolved | 11 |
| Partner coordinate unresolved | 6 |

The 54 Paulin Paz v2 records and 134 Valle-Inclan lifted records remain separate benchmarks. Eleven VCF records have opposite-mate context, with no observed same-read split candidate in the retained BAM. No junction or SV-derived protein is called.

## Sources, processing and limits

| Input | Pinned scope |
|---|---|
| Matched-normal small variants | SMaHT COLO829/COLO829BL, GRCh38 no-alt |
| SV benchmarks | Paulin Paz 2024 v2 (54 records) and Valle-Inclan 2022 lifted (134 records) |
| Original RNA | SRR8615617; 115,269,780 paired Illumina spots / 230,539,560 reads |
| Alignment | minimap2 2.26 -ax sr --frag=yes; NCBI GRCh38 no-alt |

## Reference model and interpretation

Small-variant coding effects use Varcode with Ensembl release 87 on GRCh38. The human orfdata 0.34.1 catalog uses Ensembl 101 GRCh38 reference ORFs (7,264 models, dataset 2026-10-11.1). An orfdata coding-block overlap is a reference annotation candidate; this analysis did not translate mutated ncORFs from the RNA reads. Genomic transcript overlap is not RNA-compatible isoform support. Original-read allele counts and local protein contexts are separate observations.

## SHA-256 input pins

**SMaHT SNV VCF gzip**

`e57f77c8bc280bbad2255856c4062348f92f25da38726f367322c52736578f31`
**SMaHT indel VCF gzip**

`e8d0fb67925a2365b485cd65feab8dfdc12bbe54c0bc0bdec44c1846b518b685`
**SV v2 GRCh38 VCF**

`8ad075bf43aa71d89c591396569e813959ada44a30f171982c2d299411290bac`
**Earlier SV lifted VCF**

`a8f38454367cf8837a7dba1d29c9bc338a8e62cadfee6d2489931f8105f0dcdc`
**SRR8615617 SRA archive**

`9d89bb4f5a74fba7a8ba9495fd44046c35f4abb60d10c7bd9261f07569520b4f`
**GRCh38 no-alt FASTA (uncompressed)**

`9cce8b926416dd96b152deea85188495b75f7ac8d634cc723a017067be8702b7`
**orfdata bundled GRCh38 catalog**

`06bc0bf3f81c25a7356f71a38581ff5704c05dcb75752454bc1ca75075ab8c09`

## What remains unresolved

The SMaHT DNA, CCLE RNA and TAGET polished isoforms are separate cultures. The short-read genomic alignment is not splice-aware; junction-crossing reads may be lost. Full-length isoform linkage, SV junction paths and SV-derived proteins were not resolved.

Machine-readable per-call annotation, allele, protein, and SV tables and complete reconstruction commands are published with the Isovar HCC1395/COLO829 reports. These PDFs summarize selected public source call sets, not a complete catalogue of every cell-line mutation.